

Configuring an IPSec Remote Desktop VPN on FortiGate-40f



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Purpose:

The purpose of this lab is to configure a remote-access (dial-up) IPSec VPN on a FortiGate 40F so that an off-site user running the FortiClient software can establish a secure, encrypted tunnel into the local network over the public internet. Once connected, the remote user authenticates with their own credentials and a shared key tied to a fixed identifier, and is then able to securely reach internal resources, demonstrated in this lab by establishing a Remote Desktop session to an internal host across the tunnel.

Background Information:

IPSec (Internet Protocol Security) is a suite of protocols that provides confidentiality, integrity, and authentication for traffic traversing an untrusted network such as the internet. Rather than relying on a single mechanism, IPSec negotiates a secure connection in two distinct phases using the Internet Key Exchange (IKE) protocol. Phase 1 establishes a secure, authenticated management channel (the IKE Security Association) between the two endpoints, during which the peers agree on encryption and hashing algorithms, a Diffie-Hellman group for key exchange, and a method of mutual authentication. Phase 2 then uses that protected channel to negotiate with the IPSec Security Associations that encrypt and encapsulate user data, again selecting agreed-upon encryption and authentication proposals. Because both endpoints must agree on every one of these parameters, the proposals configured on the FortiGate and the FortiClient must match exactly, or the tunnel will fail to come up.

VPNs are generally deployed in one of two models. A site-to-site VPN permanently links two fixed networks, such as two branch offices, while a remote-access or "dial-up" VPN allows individual roaming clients to connect into a central network from arbitrary, often unknown, public IP addresses. This lab implements the dial-up model, which is why the FortiGate is configured to accept incoming connections from clients rather than to a single predefined peer address. Authentication in this lab is handled using a pre-shared key (PSK), a secret string configured identically on both ends. By default, the FortiGate dial-up configuration will accept any peer ID, meaning any client presenting the correct pre-shared key is allowed to negotiate. To tighten this, the lab configures Phase 1 to require a specific Peer ID and sets the Local ID to that same value, so that the firewall only accepts clients that present the expected identifier in addition to the correct key. This is a meaningful security improvement because it adds a second matching

condition beyond the shared secret and helps prevent unauthorized clients from even beginning a negotiation. User authentication is layered on top of the tunnel through a local user definition and a user group on the FortiGate. When a client connects, in addition to the pre-shared key and matching peer ID, the user must supply valid credentials, providing per-user accountability rather than relying solely on a single shared secret. This is conceptually similar to the RADIUS-backed authentication used in the wireless lab, except that the user database here is stored locally on the firewall itself.

A critical operational consideration for IPSec is the network path between the client and the firewall. IKE negotiation occurs over UDP port 500, and when either endpoint sits behind a NAT device, NAT-Traversal (NAT-T) shifts the encrypted traffic to UDP port 4500. If an upstream router or ISP blocks either of these ports, the tunnel cannot be established even when every setting is correct, which is one of the most common causes of dial-up VPN failures. Finally, the FortiClient application serves as the software IPSec client on the remote machine; every parameter inside it, the Phase 1 Local ID, the Phase 1 and Phase 2 encryption proposals, the encapsulation mode, and the pre-shared key, must mirror the FortiGate's configuration for the connection to succeed.

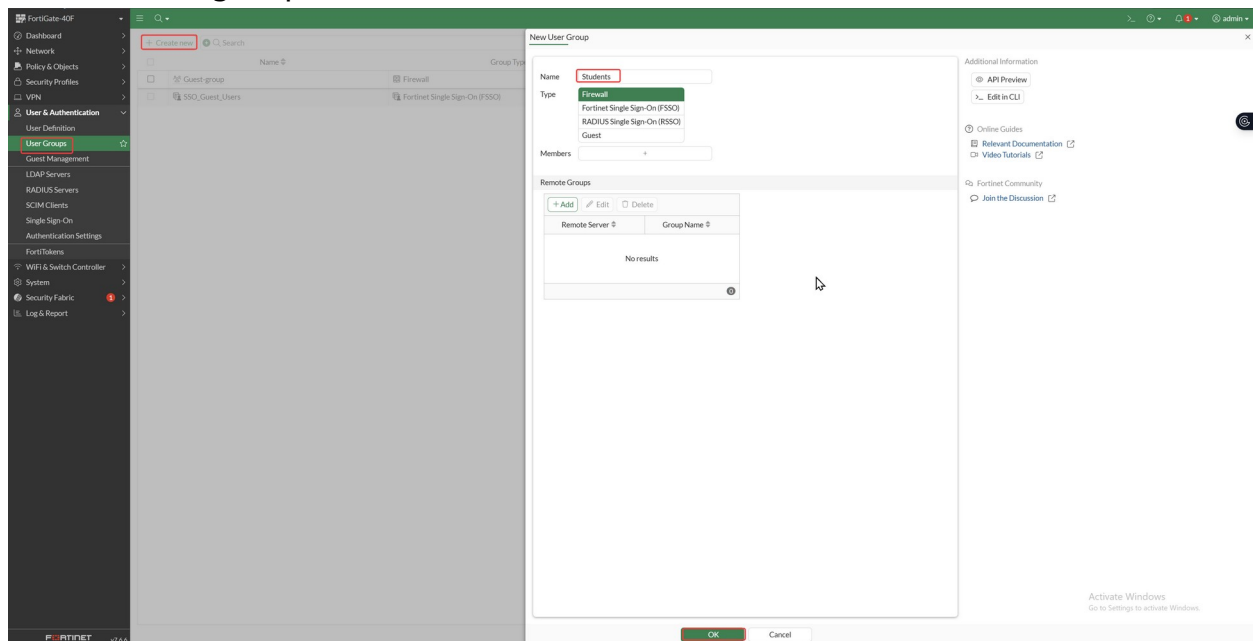
Lab Summary:

In this lab, we configured a remote-access IPSec VPN on a FortiGate 40F and connected to it from a remote machine using FortiClient, then used the secured tunnel to reach an internal host over Remote Desktop. We began on the FortiGate by creating a user group and a local user definition that the VPN would authenticate against, skipping the optional two-factor authentication. We then used the built-in VPN Wizard with its default dial-up template to create the IPSec VPN, which automatically generated the corresponding Phase 1, Phase 2, firewall, and routing settings. After the wizard completed, we returned to the VPN configuration and edited the authentication settings so that, instead of accepting any peer ID, the tunnel required a specific pre-set Peer ID, and we set the Phase 1 Local ID to that same value to harden the negotiation. On the client side, we downloaded and installed FortiClient from Fortinet's official download page and added a new IPSec VPN connection, carefully matching every parameter to the FortiGate's configuration, including the Phase 1 Local ID, the Phase 1 and Phase 2 encryption proposals, the encapsulation method, and the pre-shared key, while leaving Enable Local LAN unchecked in Phase 1. We then

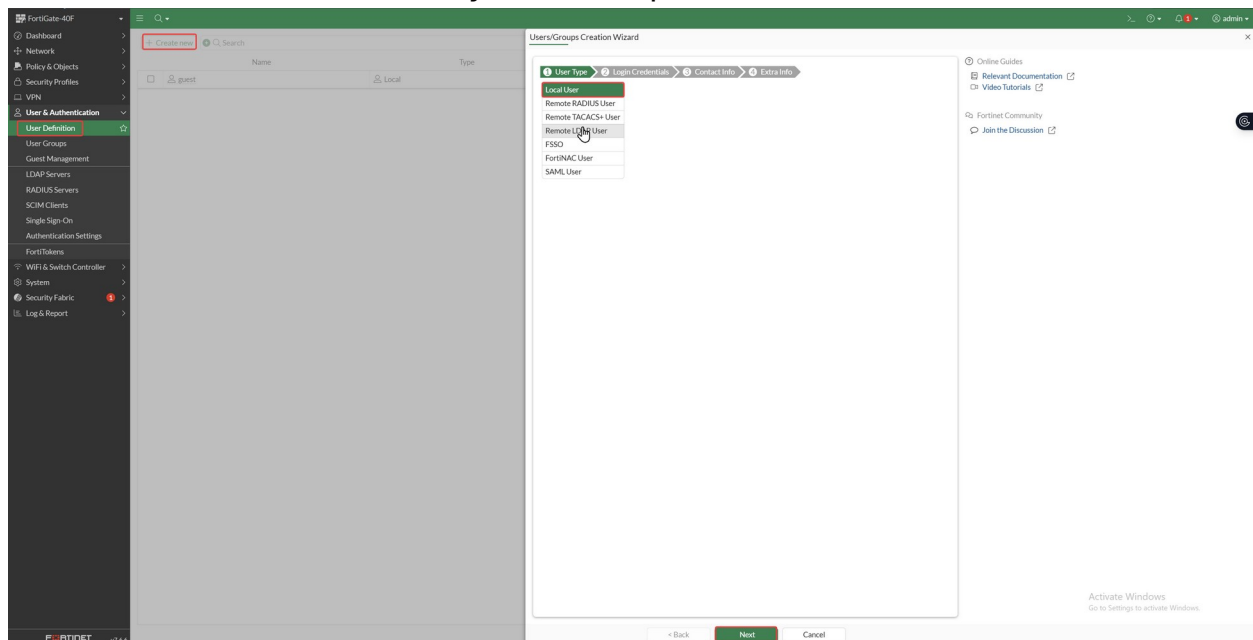
connected to the VPN, and where connection failures occurred, we verified that UDP ports 500 and 4500 were not being blocked by the router or ISP, that the FortiClient settings exactly matched the firewall, and that the FortiGate was not blocking traffic from the subnet or ports in use. With the tunnel established, we opened Remote Desktop Connection and authenticated to an internal host across the VPN, and captured a packet capture as proof that the traffic traversing the tunnel was encrypted and secure.

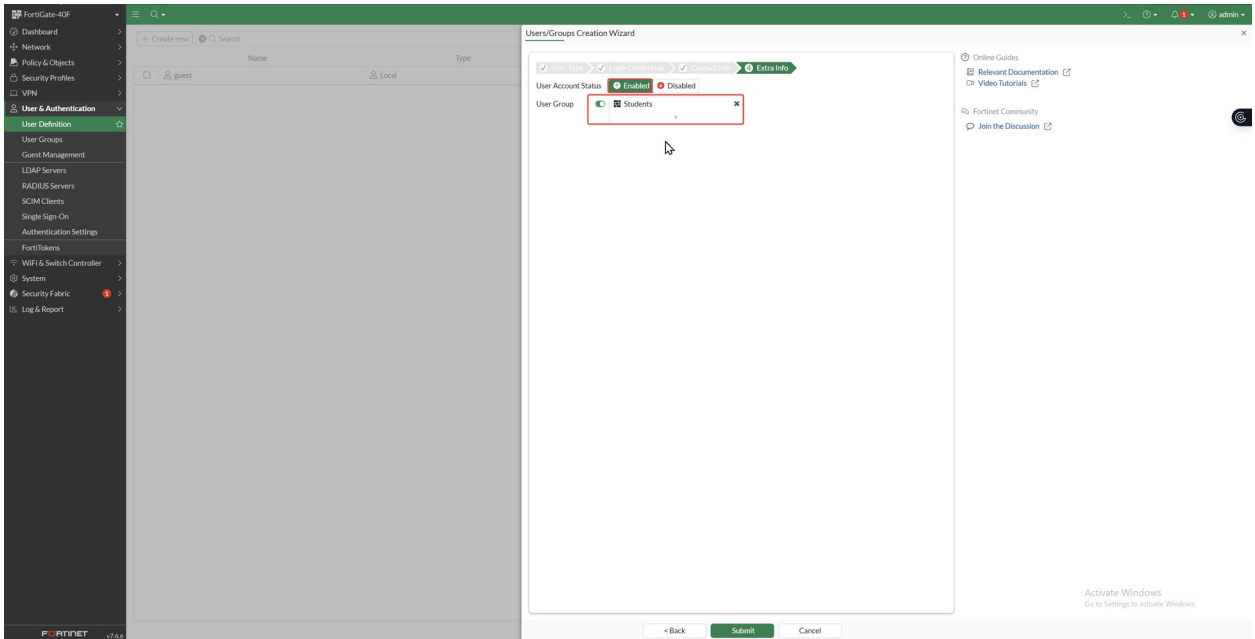
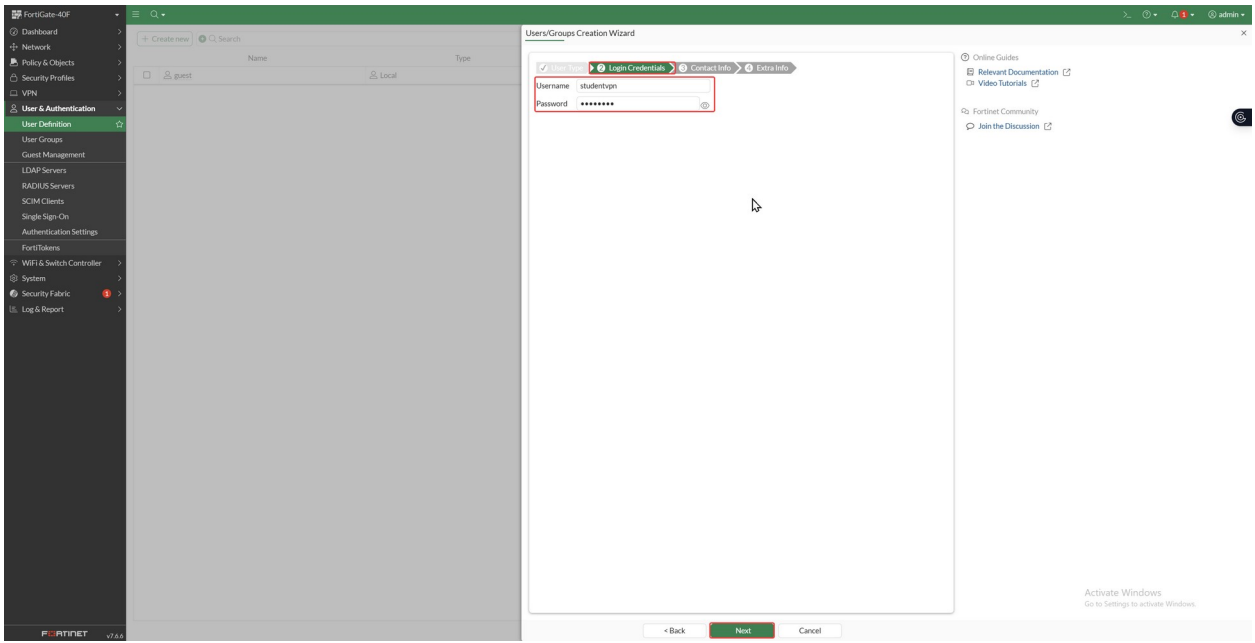
Steps and Configuration:

Create a user group inside the Firewall:



Create a local user definition (you can skip the 2FA)





Create a VPN with the default VPN Wizard template:

The screenshot shows the FortiGate VPN Wizard interface. The left sidebar contains the navigation menu with 'VPN' and 'VPN Wizard' highlighted. The main content area displays the 'VPN Wizard' title and a introductory text box. Below this, the 'Tunnel name' field is set to 'RemoteAccessVPN'. Under 'Select a template', three options are shown: 'Remote Site', 'VPN Tunnel', and 'Local FortiGate'. The 'VPN Tunnel' option is selected. Below this, two sub-templates are shown: 'Site to Site' and 'Hub and Spoke with ADVPN'. The 'Remote Access' sub-template is highlighted with a red box. It shows a diagram with 'Remote Endpoints', 'VPN Tunnels', and 'Local FortiGate'. The 'Begin' button is at the bottom right.

FortiGate-40F

Dashboard

Network

Policy & Objects

Security Profiles

VPN

VPN Tunnels

VPN Wizard

VPN Location Map

User & Authentication

WiFi & Switch Controller

System

Security Fabric

Log & Report

VPN Wizard

The IPsec VPN tunnel wizard gets you started with templates for common use cases. If the templates below do not fulfill your needs, fully custom tunnels can be set up. For convenience, it is also possible to start with a template and convert it to a fully custom tunnel after.

Tunnel name: RemoteAccessVPN

Select a template

Remote Site

VPN Tunnel

Local FortiGate

Site to Site

Offices in multiple, fixed locations can establish secure connections with each other. Branch offices can also access the main office's intranet.

Supported remote peers:

Hub and Spoke with ADVPN

VPN connections radiate from a central FortiGate unit (the hub) to multiple remote peers (the spokes). Traffic can pass between private networks that are each behind a hub or remote peer (spoke). Additionally, spoke to spoke traffic is possible via ADVPN tunnels.

Supported remote peers:

Remote Access

Employees who need to access their company's network from off-site locations or users who want to securely connect to a private network from a public area frequently use this type of VPN.

Begin

The screenshot shows the 'VPN Wizard - RemoteAccessVPN' configuration page. At the top, a diagram shows 'Remote Endpoint', 'VPN Tunnel', and 'Local FortiGate'. The 'VPN tunnel' section contains the following configuration:

- VPN client type: ☒ Windows, ☐ Apple, ☐ Android, ☐ Linux, ☐ Cisco
- Authentication method: ☒ Pre-shared key, ☐ Signature
- Pre-shared key:
- IKE: ☒ Version 2, ☐ Version 1
- Transport: ☐ UDP, ☒ Auto
- Use Fortinet encapsulation: ☒
- NAT traversal: ☒ Enable, ☐ Disable, ☐ Forced
- Keepalive frequency: 10
- EAP peer identification: ☐ IKEv2 IDi payload, ☒ EAP identity request
- User authentication method: ☒ Phase 1 interface, ☐ Inherit from policy
- DNS Server: ☐ Use System DNS, ☒ Specify

The 'Specify' button for DNS Server is highlighted with a red box. The 'Next' and 'Cancel' buttons are at the bottom.

VPN Wizard - RemoteAccessVPN

Remote Endpoint

VPN Tunnel

Local FortiGate

VPN tunnel

VPN client type

Authentication method

Pre-shared key

IKE

Transport

Use Fortinet encapsulation

NAT traversal

Keepalive frequency

EAP peer identification

User authentication method

DNS Server

Next

Cancel

VPN Wizard - RemoteAccessVPN

Remote Endpoint

VPN Tunnel

IncomingLocal FortiGateInternal

VPN tunnel

Remote endpoint

Addresses to assign to connected endpoints

10.1.2.3-10.1.2.250

Subnet for connected endpoints

255.255.255.255

FortiClient settings

Security posture gateway matching

EMS SN verification

Save password

Auto Connect

Always up (keep alive)

NextCancel

VPN Wizard - RemoteAccessVPN

Remote Endpoint

VPN Tunnel

IncomingLocal FortiGateInternal

VPN tunnel

Remote endpoint

Local FortiGate

Incoming interface that binds to tunnel

wan

Create and add interface to zone

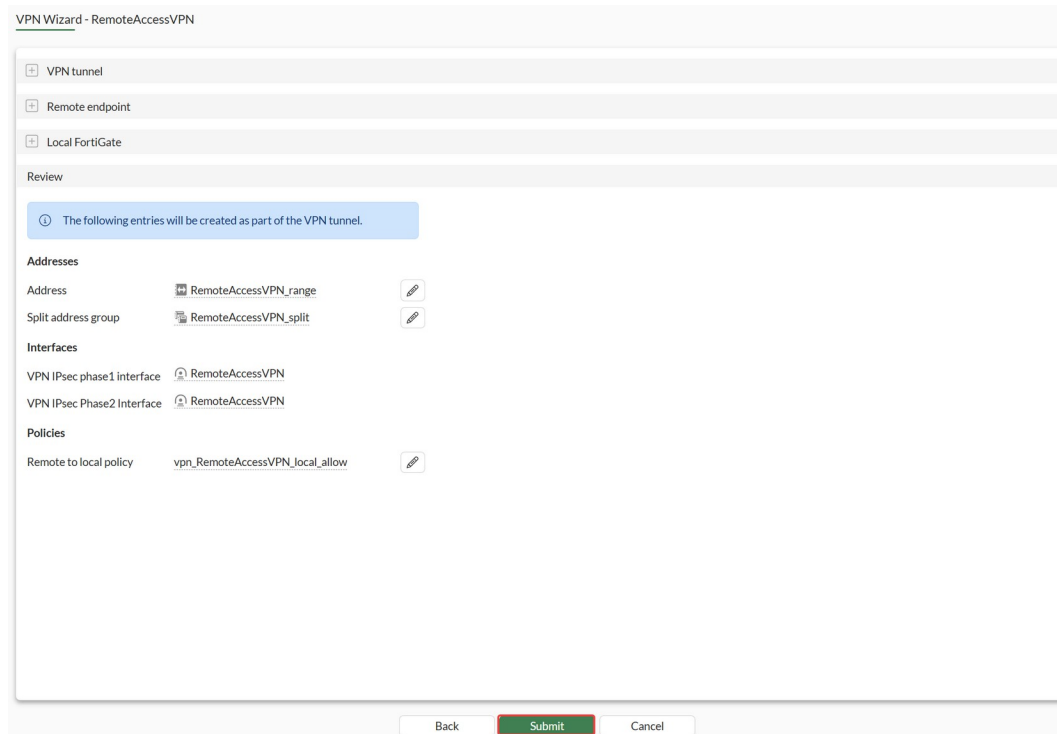
Local interface

Soho LAN

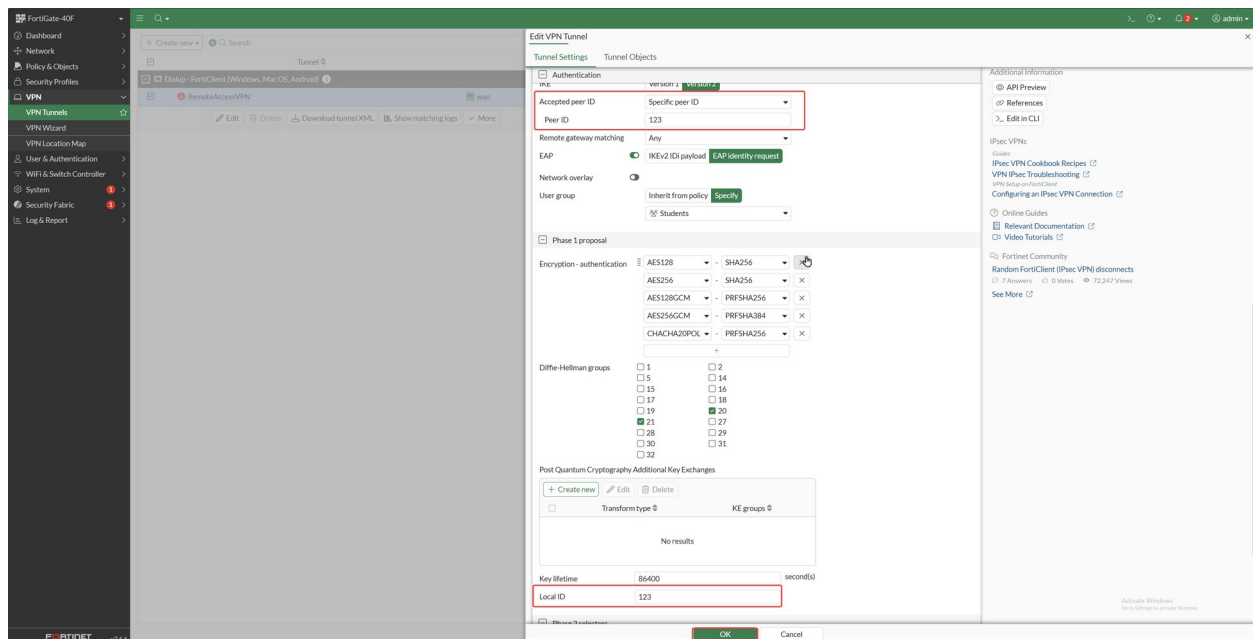
Local Address

Soho LAN address

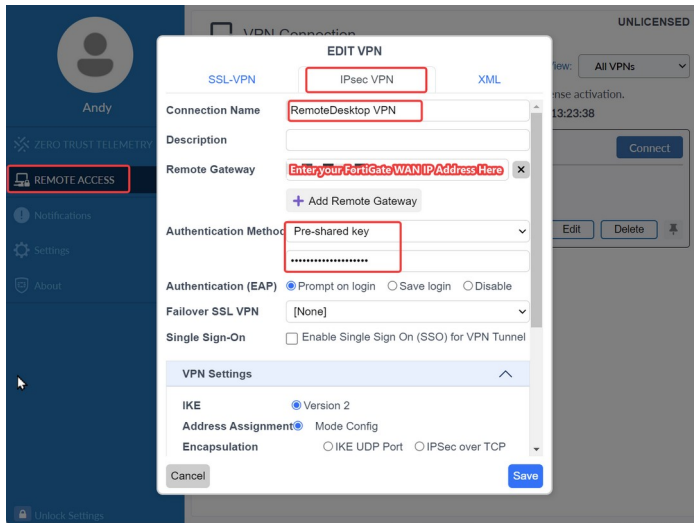
NextCancel



Now go back into the VPN Setup and Edit the authentication settings to set a pre-set Peer ID instead of allowing any peer ID. Also change the local ID in Phase 1 to the same as the Peer ID.



Now install FortiClient on their official website at <https://www.fortinet.com/support/product-downloads> and add a new VPN:



VPN Settings

IKE ☒ Version 2

Address Assignment ☒ Mode Config

Encapsulation ☐ IKE UDP Port ☐ IPsec over TCP

☒ Auto (UDP fallback TCP)

IKE UDP Port

IKE TCP Port

Phase 1

IKE Proposal

Encryption: AES256

Authentication: SHA256

Encryption: AES256

Authentication: SHA256

DH Group

☐ 1 ☐ 2 ☐ 5 ☐ 14 ☐ 15

☐ 16 ☐ 17 ☐ 18 ☐ 19 ☒ 20

☒ 21 ☐ 27 ☐ 28 ☐ 29 ☐ 30

☐ 31 ☐ 32

Key Life Seconds

Local ID

☒ Dead Peer Detection

Phase 2

IKE Proposal

Encryption: AES256

Authentication: SHA256

Encryption: AES256

Authentication: SHA256

Key Life ☒ 43200 Seconds

☐ 5120 KBytes

☒ Enable Replay Detection

☒ Enable Perfect Forward Secrecy (PFS)

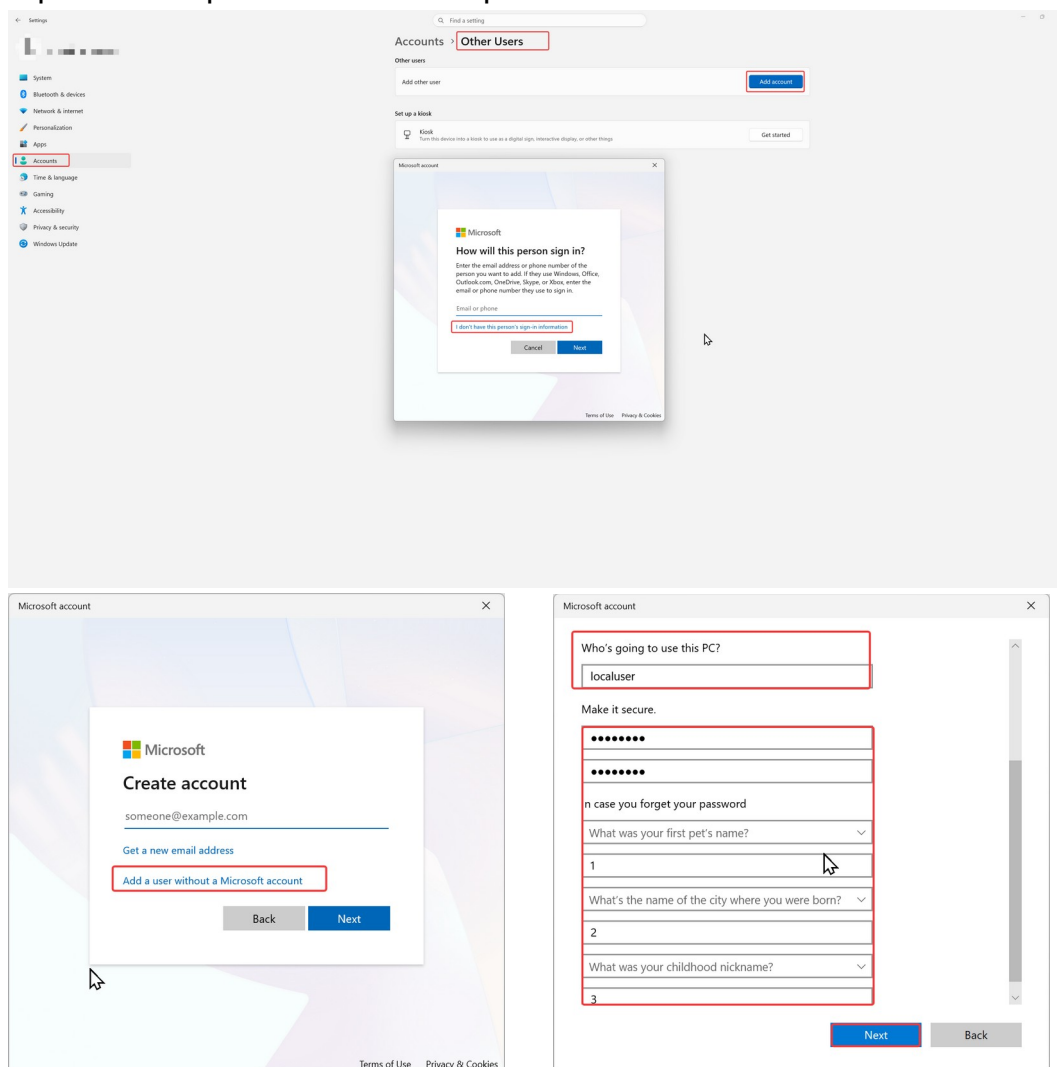
DH Group

Note: Leave Enable Local LAN unchecked in Phase 1.

Now press Connect and connect to your VPN! If it fails to connect, check the following information.

1. Your router or ISP blocks port 500 or port 4500. If these UDP ports are blocked, the VPN will not work.
2. Ensure that all information inside the FortiClient matches that of the FortiGate Firewall, most importantly the Phase 1 Local ID, encryption methods inside proposal for Phase 1 and 2, encapsulation in VPN Settings, and that the Pre-Shared Key was typed in correctly.
3. The FortiGate isn't blocking traffic from the subnet or ports in use.

Steps to set up Remote Desktop:



Accounts > Other Users

Other users

Add other user

Add account

localuser
Local account

Account options

Change account type

Account and data

Remove

Change account type

localuser
Local account

Administrator

Standard user

OK

Cancel

System > Remote Desktop



Remote Desktop

Connect to and use this PC from another device using the Remote Desktop app

On



PC name

Use this name to connect to this PC from another device



Remote Desktop users

Select who can remotely access this PC



Remote Desktop Users

The users listed below can connect to this computer, and any members of the Administrators group can connect even if they are not listed.

☐ already has access.

Add...

Remove

To create new user accounts or add users to other groups, go to Control Panel and open [User Accounts](#).

OK

Cancel

Select Users

Select this object type:

Users or Built-in security principals

Object Types...

From this location:



Locations...

Enter the object names to select (examples):

localuser

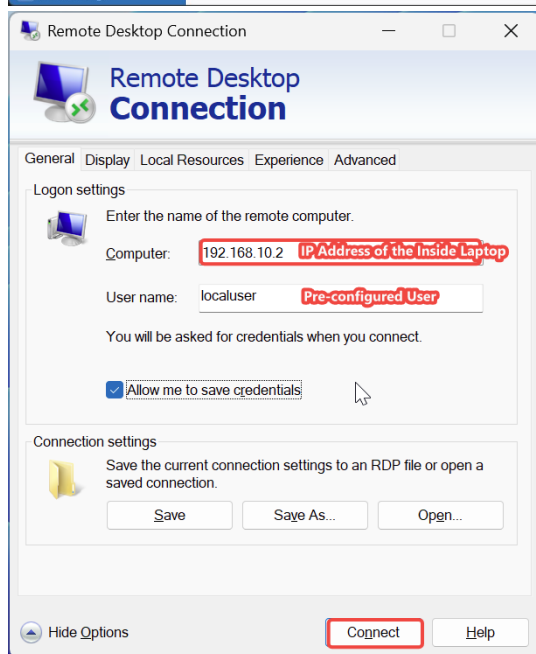
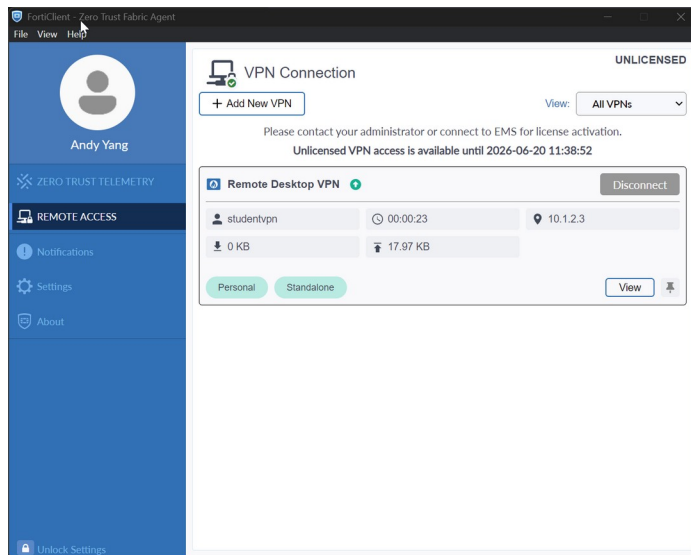
Check Names

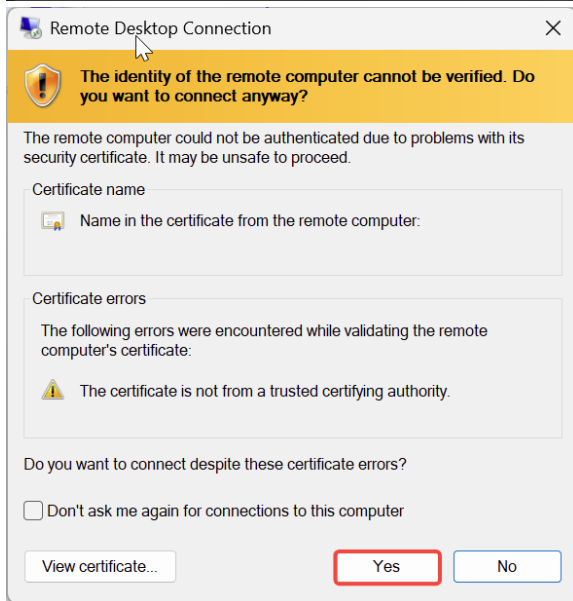
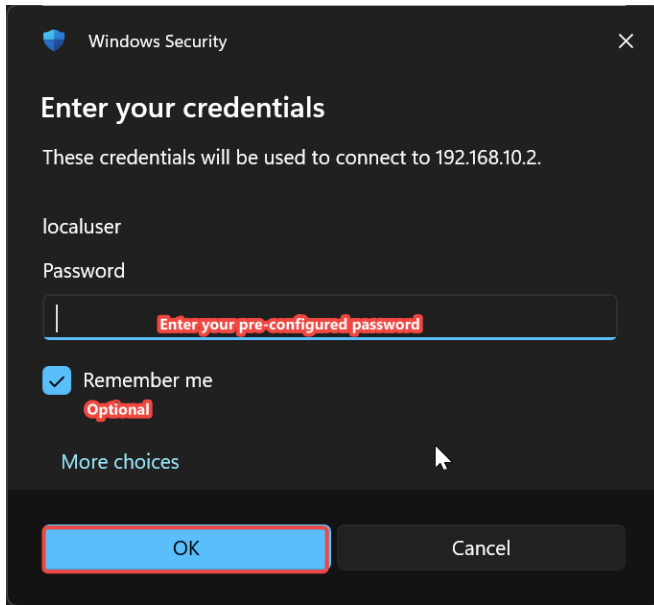
Advanced...

OK

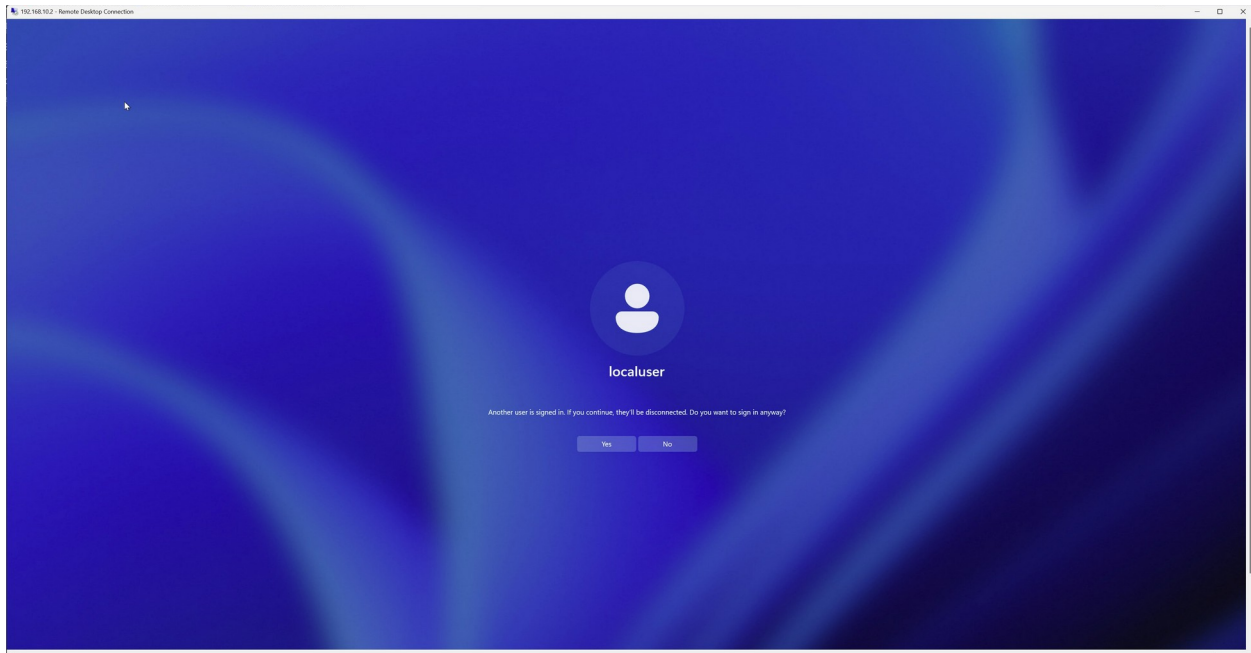
Cancel

Connect to the VPN, then open Remote Desktop Connection and enter the credentials.

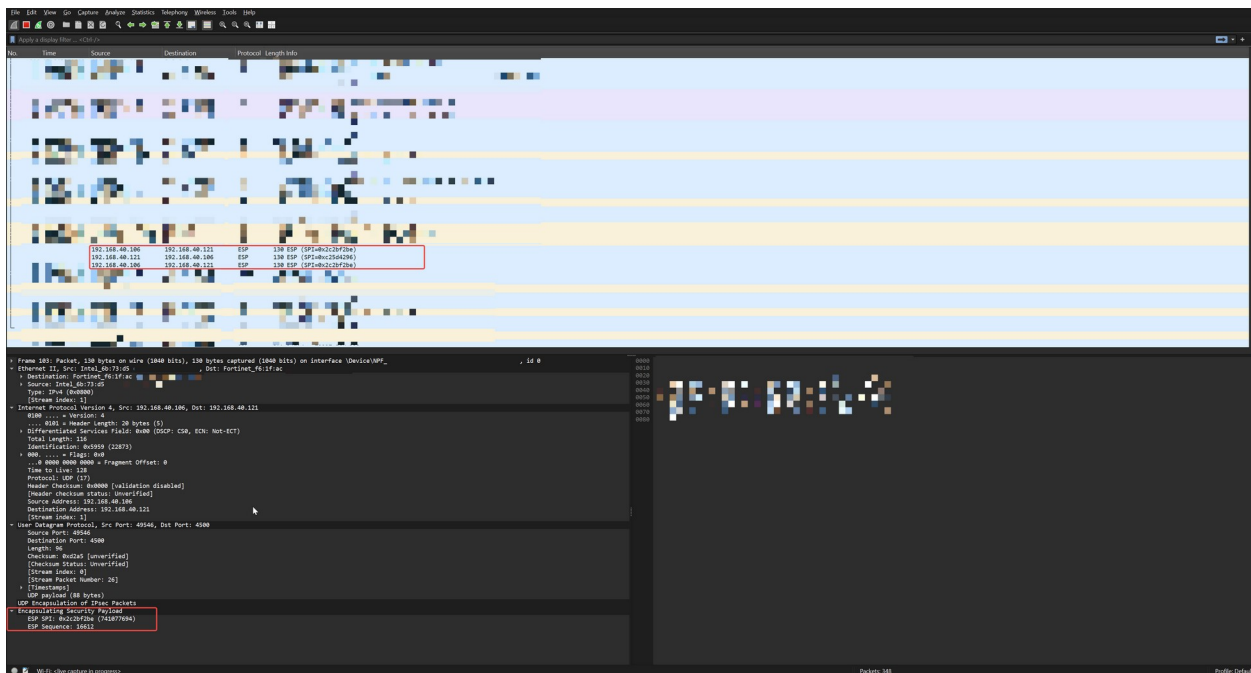




Done:



Proof of Secure Packet



Problems and Solutions:

The main problems encountered involved the Phase 1 and Phase 2 settings on the FortiClient side being misconfigured. The most confusing issue was that FortiClient has two separate places to enter a Local ID, and only the Phase 1 Local ID is the one that must match the Peer ID configured on the FortiGate 40F. Entering the ID in the wrong field produced a connection failure with no indication of the cause. Because a failed IKE negotiation gives

almost no feedback, troubleshooting required methodically comparing every parameter on both ends rather than looking for a specific error message.

Conclusion:

In this lab, we successfully configured IPSec Remote Desktop VPN on a FortiGate 40f Firewall. We utilized the VPN Wizard provided within the FortiGate 40f and then manually configured a custom local ID to add an extra layer of security so that unauthorized personnel cannot access the VPN and therefore have access to the local network.

After configuring the VPN on both server and client side, we connected to the VPN from a device outside the local network and confirmed that the tunnel was established successfully, then used Windows Remote Desktop from the external device to connect to a machine inside the local network, and the firewall's web GUI and CLI were both able to be accessed successfully through this connection, confirming that the VPN was correctly routing traffic and that internal resources were fully reachable through the encrypted tunnel as intended.